

Unifying Special Relativity and Quantum Mechanics

Lecture 17 — Linearized Gravity and the Weinberg Soft Theorem

(The Massless Limit, Coupling to $T^{\mu\nu}$, the Belinfante–Rosenfeld Tensor, and the Universality of Gravity)

Typed version of the handwritten notes, with added explanations in blue boxes

How to read this document

Conventions. Exactly as in Lectures 1–16: everything in the normal text and in the **yellow** / **pink** / **grey** boxes is a faithful transcription of the handwritten notes (yellow boxes for the yellow-highlighted results, pink boxes for the pink/red-framed ones, grey boxes for the plain pencil frames). The colours of the text roughly follow the colours of the pens used in the original. Everything inside a **blue box like this one** is extra material that I added: derivations spelled out, terminology, corrections of transcription slips, redrawn figures, and connections to earlier lectures.

Pagination of this scan. These sheets *continue Lecture 16 without interruption*: no restart of the numbering, so they are still marked with **nine stars**, running [p. 11*****]–[p. 28*****]. One cosmetic quirk: on the fourth sheet the “1” of 14 is written over the circle, so the page number reads like “④”. It is 14*****.

The scan is open at both ends.

- It *opens* mid-sentence: the first line, “...less limit”, is the tail of the sentence that Lecture 16 broke off at — “the mass term ... cannot survive the *massless limit*”. Section 1 restores the full sentence.
- It *closes* mid-sentence too: sheet [p. 28*****] stops at “to extract the propagator, we need...”. Section 15 transcribes the fragment and a blue box says how the sentence ends.

Everything in between is complete, in order, and free of duplicates.

Where we left off. Lecture 16 built the Fierz–Pauli Lagrangian for a *massive* spin-2 field and started the massless limit, noting (consideration 1) that the mass term is not gauge invariant. This lecture completes that limit, writes the free massless Lagrangian $\mathcal{L}_0$, and then repeats the Lecture-15 programme one spin higher: couple the field to a conserved source. The source turns out to be the energy–momentum tensor $T^{\mu\nu}$ — which must be made symmetric (Belinfante–Rosenfeld) — and the resulting theory is *linearized gravity*. The last third of the lecture derives Weinberg’s soft-graviton theorem and, from it, the **equivalence principle**: every particle couples to gravity with the same strength.

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Part I — The massless limit completed

(pp. 11*****–14*****)

1 Consideration 2) and the massless Lagrangian [p. 11*****]

... less limit

 $\hookrightarrow$ as $\frac{1}{2}M^2 V_\mu V^\mu$ is the spin-1 case

2) The pure derivative part of the Fierz–Pauli Lagrangian, on the other hand, is invariant under the gauge transformation, and survives the massless limit

 $\hookrightarrow$ as $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ in the spin-1 case

We thus write the Lagrangian

$$\mathcal{L}_0 = \frac{1}{2}(\partial_\mu h_{\nu\rho})(\partial^\mu h^{\nu\rho}) - (\partial_\mu h_{\nu\rho})(\partial^\nu h^{\mu\rho}) + (\partial_\mu h^{\mu\nu})(\partial_\nu h) - \frac{1}{2}(\partial_\mu h)(\partial^\mu h)$$

with the field $h^{\mu\nu}$ now endowed with invariance under the gauge transformation

$$h^{\mu\nu} \longrightarrow h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$$

The equation of motion are given by ★ (pag. 7*****) if we set $b = -1$ and $m^2 = 0$.

We have [p. 12*****]

$$\left(\square h^{\rho\sigma} - \partial_\mu \partial^\rho h^{\mu\sigma} - \partial_\mu \partial^\sigma h^{\mu\rho} + \partial^\rho \partial^\sigma h + g^{\rho\sigma} \partial_\mu \partial_\nu h^{\mu\nu} - g^{\rho\sigma} \square h \right) = 0$$

Added explanation (not in the original notes)

The sentence that Lecture 16 cut in half. Read together, considerations 1) and 2) say: the mass term $\frac{m^2}{2}(h^2 - h_{\mu\nu}h^{\mu\nu})$ is not invariant under a gauge transformation, and cannot survive the massless limit — as $\frac{1}{2}M^2 V_\mu V^\mu$ in the spin-1 case; the pure derivative part, on the other hand, is invariant and does survive — as $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ in the spin-1 case.

The second half is the remarkable one. That the mass term breaks gauge invariance is obvious. That the kinetic part is gauge invariant is not: the four structures have fixed relative coefficients, and those coefficients were determined back on p. 4***** by a completely different requirement — the *absence of ghosts*. Two independent demands, one answer. Exactly as in electromagnetism, where $-\frac{1}{4}F^2$ is simultaneously the unique ghost-free spin-1 kinetic term and the unique gauge-invariant one.

Getting the signs of the E.o.M. Take ★ from p. 7***** ,

$$b \left[-\square h^{\rho\sigma} + \partial_\mu \partial^\rho h^{\mu\sigma} + \partial_\mu \partial^\sigma h^{\mu\rho} - \partial^\rho \partial^\sigma h - g^{\rho\sigma} \partial_\mu \partial_\nu h^{\mu\nu} + g^{\rho\sigma} \square h \right] = 2em^2 (h^{\rho\sigma} - g^{\rho\sigma} h),$$

set $m^2 = 0$ so the right-hand side vanishes, and set $b = -1$, which flips every sign in the bracket. That is the boxed equation.

2 The wave operator $\mathcal{G}^{\rho\sigma}$ [p. 12*****]

We introduce a short-hand notation by defining

$$\mathcal{G}^{\rho\sigma}(h(x)) \equiv \square h^{\rho\sigma} - \partial_\mu \partial^\rho h^{\mu\sigma} - \partial_\mu \partial^\sigma h^{\mu\rho} + \partial^\rho \partial^\sigma h + g^{\rho\sigma} \partial_\mu \partial_\nu h^{\mu\nu} - g^{\rho\sigma} \square h$$

$$\mathcal{G}^{\rho\sigma}(h(x)) \equiv \partial_\mu \frac{\partial \mathcal{L}_0}{\partial (\partial_\mu h_{\rho\sigma})}$$

↪ so that the E.o.M. now are just given by

$$\mathcal{G}^{\rho\sigma}(h(x)) = 0$$

E.o.M. in terms of $\mathcal{G}^{\rho\sigma}(h(x))$

We note the following property

$$\partial_\rho \mathcal{G}^{\rho\sigma}(h(x)) = 0$$

This is true whatever field configuration.

Added explanation (not in the original notes)

$\mathcal{G}^{\rho\sigma}$ is (twice) the linearized Einstein tensor. Expanding the Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$ to first order around flat space with $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ gives

$$G^{(1)\rho\sigma} = \frac{1}{2} \left(\partial_\mu \partial^\rho h^{\mu\sigma} + \partial_\mu \partial^\sigma h^{\mu\rho} - \square h^{\rho\sigma} - \partial^\rho \partial^\sigma h - g^{\rho\sigma} \partial_\mu \partial_\nu h^{\mu\nu} + g^{\rho\sigma} \square h \right) = -\frac{1}{2} \mathcal{G}^{\rho\sigma}.$$

So “ $\mathcal{G}^{\rho\sigma} = 0$ ” is exactly the *linearized vacuum Einstein equation* $G_{\mu\nu}^{(1)} = 0$, and on p. 16***** the sourced version will be the linearized $G_{\mu\nu} = 8\pi G T_{\mu\nu}$. We are doing general relativity without having mentioned geometry once.

Why $\partial_\rho \mathcal{G}^{\rho\sigma} = 0$ identically. Write out the divergence: the six terms cancel in three pairs by commutativity of partial derivatives, exactly as in the computation on p. 7*****. The important word in the note is “**whatever field configuration**”: this is an *off-shell identity*, not a consequence of the equations of motion. It is the linearized **Bianchi identity** $\nabla_\mu G^{\mu\nu} \equiv 0$ of general relativity, and — as in Lecture 15 — it is an instance of Noether’s second theorem: a local (gauge) symmetry always forces such an identity among the equations of motion. Its consequence is immediate and will be used on p. 16*****: whatever couples to $h_{\mu\nu}$ must be conserved.

Notation. The original writes a script capital for this object; I typeset it as $\mathcal{G}$. It is a linear second-order differential operator acting on h , nothing more.

3 Comment: contact with the first part of the course [p. 12*****–14*****]

Let us try to make contact with what we’ve done in the first part of the course. [p. 13*****]

■ What we did before

We constructed a symmetric, traceless field $h^{\mu\nu}$ for the description of massless spin-2 particles. We found that such field is subject to the constraints $h^{\mu 0} = \partial_i h^{ij} = 0$.

↪ We then discovered that such field possesses a gauge invariance

$$h^{\mu\nu} \longrightarrow h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$$

■ What we are doing now

We constructed a Lagrangian density for a symmetric field $h^{\mu\nu}$ endowed with a gauge invariance

$$h^{\mu\nu} \longrightarrow h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$$

↪ It is possible to make a gauge choice such that the field $h^{\mu\nu}$ satisfies

$$h^{\mu 0} = \partial_i h^{ij} = h = 0$$

(a.k.a. **traceless–transverse gauge**)

This is conceptually analogue to what we have discussed in the case of electromagnetism. [p. 14*****]

■ In the first part, we constructed a field A^μ to describe massless spin-1 particles. We found that such field is subject to the conditions $A^0 = 0, \vec{\nabla} \cdot \vec{A} = 0$.

↪ We then discovered that this field has a gauge redundancy $A^\mu \rightarrow A^\mu + \partial^\mu \alpha(x)$

■ In the second part, we constructed a Lagrangian density for the description of a field A^μ with a gauge redundancy $A^\mu \rightarrow A^\mu + \partial^\mu \alpha$

↪ It is possible to make a choice of gauge such that $A^0 = 0, \vec{\nabla} \cdot \vec{A} = 0$ (a.k.a. **Coulomb gauge**)

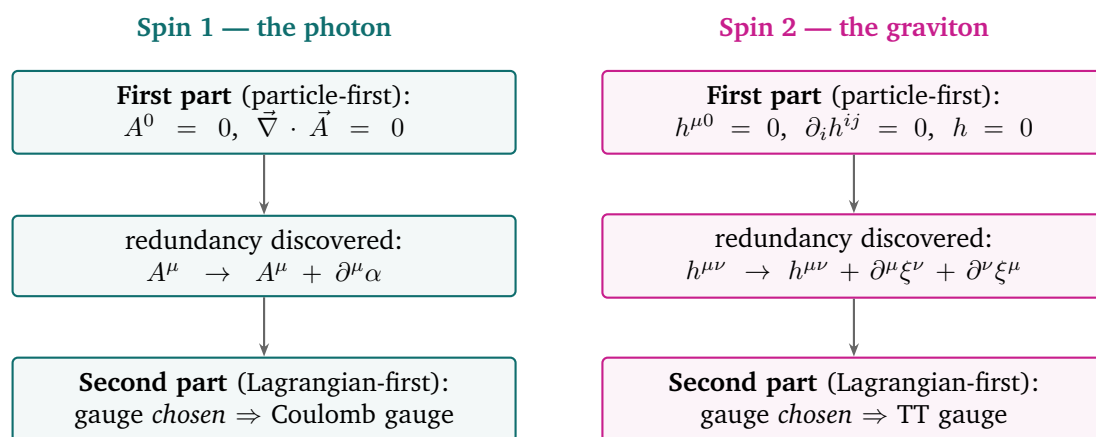


Figure 1. The two routes to the same theory, for spin 1 and spin 2. (Added figure, not in the original notes.)

Added explanation (not in the original notes)

What the comment is really pointing at. The two “parts” of the course arrived at the same place from opposite directions.

- *Part one* started from the **particle**: Wigner’s classification said a massless spin-2 particle has 2 polarizations, and forcing a ten-component field to describe only those 2 produced the conditions $h^{\mu 0} = \partial_i h^{ij} = h = 0$ — and, as a surprise, a leftover redundancy.
- *Part two* starts from the **Lagrangian**: the redundancy is now an input, and the conditions become a *choice* one is free to make (a gauge fixing), not a constraint imposed from outside.

The counting in TT gauge. $h^{\mu\nu}$ has 10 components. The gauge parameter ξ^μ has 4 components, which remove 4; the residual gauge freedom (transformations with $\square \xi^\mu = 0$ that preserve the gauge conditions) removes 4 more. $10 - 4 - 4 = 2$ — the two helicities ± 2 of the graviton. The photon case is the same story with $4 - 1 - 1 = 2$.

Why “traceless–transverse”. $h = 0$ is the traceless part; $\partial_i h^{ij} = 0$ is transversality of the spatial block; $h^{\mu 0} = 0$ removes the time components. In this gauge h^{ij} is a symmetric, traceless, transverse 3×3 matrix: $6 - 1 - 3 = 2$ independent entries, which is the cleanest way to see the two DOF.

Part II — Linearized gravity: coupling to $T^{\mu\nu}$

(pp. 14*****–22*****)

4 Towards an interacting theory: linearized gravity [p. 14*****–15*****]

We introduce interactions. We start following the same intuition we used in the case of spin-1. [p. 15*****]

We considered the coupling

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - A_\mu J^\mu \quad \text{with } J^\mu \text{ conserved, } \partial_\mu J^\mu = 0,$$

in such a way to preserve gauge invariance.

↪ We mimic such construction, and write

$$\begin{aligned} \mathcal{L} = & \frac{1}{2}(\partial_\mu h_{\nu\rho})(\partial^\mu h^{\nu\rho}) - (\partial_\mu h_{\nu\rho})(\partial^\nu h^{\mu\rho}) + (\partial_\mu h^{\mu\nu})(\partial_\nu h) \\ & - \frac{1}{2}(\partial_\mu h)(\partial^\mu h) - \kappa h_{\mu\nu}T^{\mu\nu} \end{aligned}$$

In order for gauge redundancy to be unphysical, we need the (symmetric) rank-2 tensor $T^{\mu\nu}$ to be **conserved**

$$\begin{aligned} h_{\mu\nu}T^{\mu\nu} & \xrightarrow[\text{transformation}]{\text{under a gauge}} (h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu)T^{\mu\nu} = \\ & = h_{\mu\nu}T^{\mu\nu} + 2(\partial_\mu \xi_\nu)T^{\mu\nu} \quad \text{integrating by parts} \\ & = h_{\mu\nu}T^{\mu\nu} + \underbrace{2\partial_\mu(\xi_\nu T^{\mu\nu})}_{\text{total derivative}} - \underbrace{2\xi_\nu(\partial_\mu T^{\mu\nu})}_{\text{total derivative}} = h_{\mu\nu}T^{\mu\nu} \end{aligned}$$

On the E.o.M., it vanishes if $T^{\mu\nu}$ is conserved.

Added explanation (not in the original notes)

This is Lecture 15's argument, verbatim, one spin up. There, gauge invariance of $-A_\mu J^\mu$ required $\partial_\mu J^\mu = 0$; here, gauge invariance of $-\kappa h_{\mu\nu}T^{\mu\nu}$ requires $\partial_\mu T^{\mu\nu} = 0$. The factor 2 in the second line uses the symmetry of $T^{\mu\nu}$: $(\partial_\mu \xi_\nu + \partial_\nu \xi_\mu)T^{\mu\nu} = 2(\partial_\mu \xi_\nu)T^{\mu\nu}$ *only if* $T^{\mu\nu} = T^{\nu\mu}$. So symmetry is not a decoration — it is used right here, and comment 2) on p. 17***** will have to work to guarantee it.

Why the source has to be $T^{\mu\nu}$. Two requirements pin it down: it must be a symmetric rank-2 tensor (to contract with $h_{\mu\nu}$) and it must be identically conserved. In a Lorentz invariant field theory there is exactly one such object supplied by Noether's theorem for spacetime translations — the energy–momentum tensor. Gravity couples to energy and momentum not by assumption but because nothing else is available.

5 The energy–momentum tensor as a source [p. 16*****]

We identify $T^{\mu\nu}$ with the **(canonical) energy momentum tensor**.

The E.o.M. of the theory are given by

$$\mathcal{L} = \mathcal{L}_0 - \kappa h_{\mu\nu}T^{\mu\nu} \quad \xrightarrow{\text{E.o.M.}} \quad \frac{\partial \mathcal{L}}{\partial h_{\rho\sigma}} = \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu h_{\rho\sigma})}$$

$$\frac{\partial \mathcal{L}}{\partial h_{\rho\sigma}} = -\kappa T^{\rho\sigma} ; \quad \partial_\mu \frac{\partial \mathcal{L}_0}{\partial (\partial_\mu h_{\rho\sigma})} = \mathcal{G}^{\rho\sigma}(h)$$

$$\mathcal{G}^{\rho\sigma}(h) = -\kappa T^{\rho\sigma}$$

same object defined before for the free theory

The energy–momentum tensor acts as a **source** for the field $h_{\mu\nu}$

We pause for a moment since this construction deserves few comments.

1) Since $\mathcal{G}^{\rho\sigma}$ has vanishing four-divergence, we find from the E.o.M.

$$\partial_\rho \mathcal{G}^{\rho\sigma} = -\kappa \partial_\rho T^{\rho\sigma} \rightsquigarrow 0 = \kappa \partial_\rho T^{\rho\sigma} \rightsquigarrow \partial_\rho T^{\rho\sigma} = 0$$

so that $T^{\rho\sigma}$ must be conserved. [p. 17*****]

Added explanation (not in the original notes)

This is the linearized Einstein equation. Using $\mathcal{G}^{\rho\sigma} = -2G^{(1)\rho\sigma}$ from the blue box on p. 12*****, the boxed E.o.M. reads

$$G^{(1)\rho\sigma} = \frac{\kappa}{2} T^{\rho\sigma},$$

which is $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ to first order in h , with $\kappa = 16\pi G$. Newton's constant enters as a *coupling constant*, exactly like q in QED.

Comment 1) is a consistency theorem, not an assumption. On p. 15*****, conservation of $T^{\mu\nu}$ was *imposed* so that the gauge redundancy stays unphysical. Here it comes back out of the equations of motion, because the left-hand side is identically divergence-free. The two facts are the same fact seen from two sides — and this is the spin-2 version of the QED statement $\partial_\nu F^{\nu\mu} = J^\mu \Rightarrow \partial_\mu J^\mu = 0$ from Lecture 15.

A warning that will matter later. $T^{\mu\nu}$ here is the energy–momentum tensor of the *matter* fields only. But $h_{\mu\nu}$ itself carries energy, so at the next order in κ the graviton must also source itself. Iterating this requirement is one classic route to the full nonlinear Einstein equations — gravity is nonlinear because its own field energy gravitates.

6 Comment 2): the source must be symmetric [p. 17*****–18*****]

2) The tensor $T^{\mu\nu}$ coupled to $h^{\mu\nu}$ must be **symmetric**

The canonical energy momentum tensor is given by the formula

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi^a)} (\partial^\nu \Phi^a) - g^{\mu\nu} \mathcal{L} \quad [T^{\mu\nu}] = [M]^4$$

It is **not guaranteed** that this term is symmetric

This term is obviously symmetric in $\mu \leftrightarrow \nu$

There is a simple argument to show that $T^{\mu\nu}$ is not always symmetric. Consider the Noether currents associated to Lorentz invariance

$$M^{\mu\rho\sigma} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^a)} (-i)(J^{\rho\sigma})^a_b \Phi^b + (x^\rho T^{\mu\sigma} - x^\sigma T^{\mu\rho})$$

↪ generators of the Lorentz group in the representation under which the fields transform

We compute explicitly the four-divergence [p. 18*****]

$$M^{\mu\rho\sigma} = S^{\mu\rho\sigma} + x^\rho T^{\mu\sigma} - x^\sigma T^{\mu\rho}$$

$$\hookrightarrow S^{\mu\rho\sigma} \equiv \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^a)} (-i)(J^{\rho\sigma})^a_b \Phi^b$$

$$\partial_\mu M^{\mu\rho\sigma} = \partial_\mu S^{\mu\rho\sigma} + \delta_\mu^\rho T^{\mu\sigma} + x^\rho (\partial_\mu T^{\mu\sigma}) - \delta_\mu^\sigma T^{\mu\rho} - x^\sigma (\partial_\mu T^{\mu\rho}) \stackrel{!}{=} 0$$

conservation of $T^{\mu\nu}$

since $M^{\mu\rho\sigma}$ is conserved

$$\partial_\mu S^{\mu\rho\sigma} + T^{\rho\sigma} - T^{\sigma\rho} \stackrel{!}{=} 0$$

↪ We thus get

$$\partial_\mu S^{\mu\rho\sigma} = T^{\sigma\rho} - T^{\rho\sigma}$$

This equation is telling us that if the field transforms non-trivially under Lorentz (that is, it describes spinning particles) then the canonical energy-momentum tensor is not symmetric.

What do we do in these cases?

It is possible to make $T^{\mu\nu}$ symmetric (and still conserved) via a “symmetrization procedure”

↪ This possibility follows from the fact that we have a freedom in the definition of the Noether currents.

[p. 19*****]

Added explanation (not in the original notes)

Reading $M^{\mu\rho\sigma}$. This is the Noether current of Lorentz invariance: the conserved charges $\int d^3x M^{0\rho\sigma}$ are the six generators of boosts and rotations, i.e. *total* angular momentum. It splits into two pieces:

- $x^\rho T^{\mu\sigma} - x^\sigma T^{\mu\rho}$ — the **orbital** part, the field-theory version of $\vec{L} = \vec{x} \times \vec{p}$;
- $S^{\mu\rho\sigma}$ — the **spin** part, which exists only when the field carries Lorentz indices, i.e. when $(J^{\rho\sigma})^a_b \neq 0$.

A scalar field has $(J^{\rho\sigma}) = 0$, hence $S = 0$, hence by the boxed relation $T^{\mu\nu}$ is automatically symmetric. A Dirac or vector field has $S \neq 0$, and then $T^{\mu\nu}$ generally is not.

The physical content of $\partial_\mu S^{\mu\rho\sigma} = T^{\sigma\rho} - T^{\rho\sigma}$. Spin is not separately conserved — it can be converted into orbital angular momentum — and the rate at which that happens is measured by the antisymmetric part of $T^{\mu\nu}$. Only the *total* is conserved.

Why the two cancelled terms vanish. $\partial_\mu T^{\mu\sigma} = 0$ was established in comment 1). And $\delta_\mu^\rho T^{\mu\sigma} = T^{\rho\sigma}$, which is where the antisymmetric combination comes from.

7 The Belinfante–Rosenfeld symmetrization [p. 19*****–21*****]

Let us discuss this point directly in the case of $T^{\mu\nu}$.

Consider, instead of $T^{\mu\nu}$, the object

$$\Theta^{\mu\nu} = T^{\mu\nu} + \partial_\lambda K^{\lambda\mu\nu}, \quad K^{\lambda\mu\nu} = -K^{\mu\lambda\nu}$$

$\hookrightarrow K^{\lambda\mu\nu}$ is an arbitrary rank-3 tensor which is **antisymmetric in the first two indices**

We observe that

i) $\Theta^{\mu\nu}$ is conserved

$$\partial_\mu \Theta^{\mu\nu} = \underbrace{\partial_\mu T^{\mu\nu}}_{=0} + \underbrace{\partial_\mu \partial_\lambda K^{\lambda\mu\nu}}_{=0 \text{ because of antisymmetry}} = 0$$

$T^{\mu\nu}$ is conserved

ii) $\Theta^{\mu\nu}$ leads to the **same conserved charges** defined by $T^{\mu\nu}$

If we compute [p. 20*****]

$$\begin{aligned} \int d^3\vec{x} \Theta^{0\nu} &= \int d^3\vec{x} (T^{0\nu} + \partial_\lambda K^{\lambda 0\nu}) = \underbrace{\int d^3\vec{x} T^{0\nu}}_{=P^\nu} + \underbrace{\int d^3\vec{x} \partial_\lambda K^{\lambda 0\nu}}_{=0} \\ &= P^\nu + \underbrace{\int d^3\vec{x} \partial_i K^{i 0\nu}}_{=0} = P^\nu \end{aligned}$$

In the sum over “ λ ”, $\lambda = 0$ vanishes and “ λ ” only takes spatial values

Surface integral at infinity: it vanishes under the assumption that at spatial infinity fields and derivatives die off fast enough.

We use this freedom to impose $\Theta^{\mu\nu} \stackrel{!}{=} \Theta^{\nu\mu}$

$$\begin{aligned} \Theta^{\mu\nu} &= T^{\mu\nu} + \partial_\lambda K^{\lambda\mu\nu} \stackrel{!}{=} \Theta^{\nu\mu} = T^{\nu\mu} + \partial_\lambda K^{\lambda\nu\mu} \\ &\Downarrow \\ \partial_\lambda (K^{\lambda\mu\nu} - K^{\lambda\nu\mu}) &= T^{\nu\mu} - T^{\mu\nu} = \partial_\lambda S^{\lambda\mu\nu} \end{aligned}$$

We thus find

$$K^{\lambda\mu\nu} - K^{\lambda\nu\mu} = S^{\lambda\mu\nu} \quad (1)$$

This equation can be solved as follows. We consider cyclic permutation of indices:

► permutation $(\lambda\mu\nu) \rightarrow (\mu\nu\lambda)$

$$K^{\mu\nu\lambda} - K^{\mu\lambda\nu} = S^{\mu\nu\lambda} \quad (2)$$

► permutation $(\lambda\mu\nu) \rightarrow (\nu\lambda\mu)$ [p. 21*****]

$$K^{\nu\lambda\mu} - K^{\nu\mu\lambda} = S^{\nu\lambda\mu} \quad (3)$$

We take (1) + (2) − (3), and find

$$K^{\lambda\mu\nu} - K^{\lambda\nu\mu} + K^{\mu\nu\lambda} - K^{\mu\lambda\nu} - K^{\nu\lambda\mu} + K^{\nu\mu\lambda} = S^{\lambda\mu\nu} + S^{\mu\nu\lambda} - S^{\nu\lambda\mu}$$

$$\underbrace{(K^{\lambda\mu\nu} - K^{\mu\lambda\nu})}_{2K^{\lambda\mu\nu}} + \underbrace{(-K^{\lambda\nu\mu} - K^{\nu\lambda\mu})}_{=0} + \underbrace{(K^{\mu\nu\lambda} + K^{\nu\mu\lambda})}_{=0} = S^{\lambda\mu\nu} + S^{\mu\nu\lambda} - S^{\nu\lambda\mu}$$

So that

$$K^{\lambda\mu\nu} = \frac{1}{2}(S^{\lambda\mu\nu} + S^{\mu\nu\lambda} - S^{\nu\lambda\mu})$$

We find

$$\Theta^{\mu\nu} = T^{\mu\nu} + \frac{1}{2}\partial_\lambda(S^{\lambda\mu\nu} + S^{\mu\nu\lambda} - S^{\nu\lambda\mu})$$

$$S^{\lambda\mu\nu} \equiv \frac{\partial \mathcal{L}}{\partial(\partial_\lambda \Phi^a)}(-i)(J^{\mu\nu})^a_b \Phi^b$$

↪ **Belinfante–Rosenfeld energy–momentum tensor**

Added explanation (not in the original notes)

Why the improvement term is invisible. Two facts do all the work. (i) $\partial_\mu \partial_\lambda K^{\lambda\mu\nu} = 0$ because $\partial_\mu \partial_\lambda$ is symmetric in $(\mu\lambda)$ while K is antisymmetric there — so conservation survives. (ii) $\int d^3\vec{x} \partial_\lambda K^{\lambda 0\nu}$ reduces to $\int d^3\vec{x} \partial_i K^{i 0\nu}$ (the $\lambda = 0$ term is $K^{00\nu} = 0$ by antisymmetry), which is a total spatial divergence, hence a surface term at infinity — so the total energy and momentum are unchanged. *Different tensor, same physics.*

Checking the three cancellations. Using $K^{\lambda\mu\nu} = -K^{\mu\lambda\nu}$: $K^{\lambda\mu\nu} - K^{\mu\lambda\nu} = 2K^{\lambda\mu\nu}$; $-K^{\lambda\nu\mu} - K^{\nu\lambda\mu} = -K^{\lambda\nu\mu} + K^{\lambda\nu\mu} = 0$; $K^{\mu\nu\lambda} + K^{\nu\mu\lambda} = 0$. The trick of adding two cyclic permutations and subtracting the third is exactly the one used to solve for the Christoffel symbols from the metric-compatibility condition in general relativity — same algebra, different letters.

So which tensor does gravity see? The Belinfante one. It is symmetric, conserved, gives the same total P^μ , and — a fact worth knowing — it coincides with the tensor obtained by varying the matter action with respect to the metric, $T^{\mu\nu}_{\text{Belinfante}} = \frac{2}{\sqrt{-g}} \frac{\delta S_{\text{matter}}}{\delta g_{\mu\nu}}$. That is the definition used in general relativity, and it is automatically symmetric because $g_{\mu\nu}$ is. Our route arrived at the same object from flat-space Noether reasoning alone.

8 Comment 3): dimensions and non-renormalizability [p. 22*****]

It is, therefore, natural to couple $h_{\mu\nu}$ with the Belinfante–Rosenfeld energy–momentum tensor.

3) We have that

$$[T^{\mu\nu}] = [\Theta^{\mu\nu}] = [M]^4$$

Consequently, the interaction $\kappa h_{\mu\nu} T^{\mu\nu}$ is **non-renormalizable** and $[\kappa] = [M]^{-1}$.

Added explanation (not in the original notes)

The dimensional bookkeeping. $[\mathcal{L}] = [M]^4$ and $[h] = [M]$ (it has a canonically normalized kinetic term $\partial h \partial h$), so $[\kappa] + [h] + [T] = 4 \Rightarrow [\kappa] = 4 - 1 - 4 = -1$. A coupling with negative mass dimension means a **non-renormalizable** interaction (Lecture 13's classification), and it identifies the scale at which the description must break down:

$$\kappa \sim \frac{1}{M_{\text{Pl}}}, \quad M_{\text{Pl}} \sim 10^{19} \text{ GeV}.$$

What that does and does not mean. It does *not* mean the theory is wrong or useless: linearized gravity is an **effective field theory**, predictive at energies $E \ll M_{\text{Pl}}$, and it reproduces every classical test of general relativity plus calculable quantum corrections. It does mean that the theory cannot be extrapolated to arbitrarily high energy without new physics — unlike QED, whose coupling q is dimensionless. This single line of dimensional analysis is the whole “problem of quantum gravity” in one sentence.

Contrast with Lecture 15. There, the photon coupling $q A_\mu \bar{\Psi} \gamma^\mu \Psi$ had $[q] = 0$: renormalizable, and minimal coupling was the natural prescription. Here the counterpart is inevitably non-renormalizable, because the source $T^{\mu\nu}$ already saturates the dimension budget on its own.

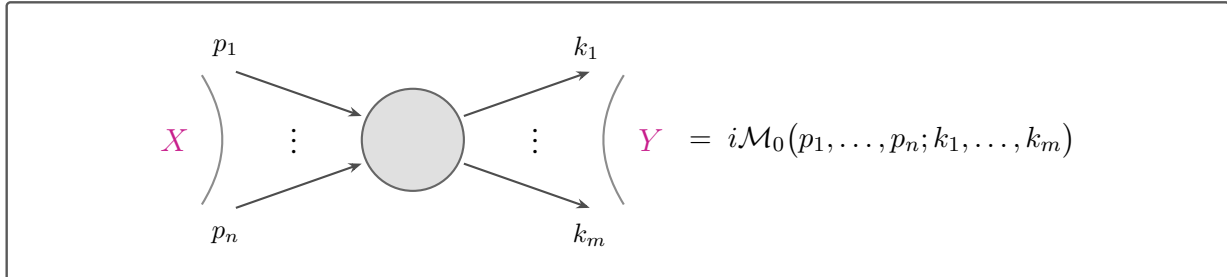
Part III — The Weinberg soft theorem for gravitons

(pp. 22*****-28*****)

9 The Weinberg soft theorem: the setup [p. 22*****-23*****]

The setup is similar to the one we discussed in the case of photons.

We thus consider the process $X \rightarrow Y$ with amplitude given by



And a process in which we have an additional graviton in the final state:

$$X \rightarrow Y + g(k)$$

There are diagrams in which the graviton is emitted by initial-state legs, final state legs, and internal lines. [p. 23*****]

Added explanation (not in the original notes)

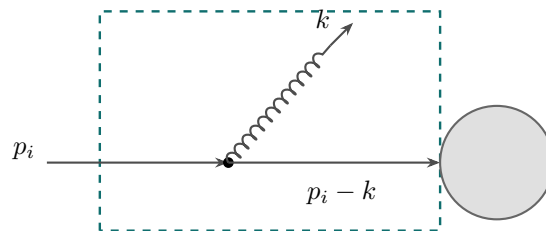
What a “soft theorem” is. It is a statement about the *universal* behaviour of an amplitude when one massless external particle has vanishing momentum $k \rightarrow 0$. The emission factors below all carry a pole $1/(p \cdot k)$, so the amplitude *diverges* as the graviton becomes soft — and the coefficient of that divergence is completely fixed by the hard amplitude $\mathcal{M}_0$ and the external momenta, with no reference to the interaction details. That is what makes the theorem powerful: whatever the unknown blob contains, the soft limit is dictated.

Why the notes did the photon case first. The photon version (Weinberg 1965) gives $\mathcal{M}_1 \approx \mathcal{M}_0 \varepsilon_\mu^* [\sum_j q_j k_j^\mu / (k_j \cdot k) - \sum_i q_i p_i^\mu / (p_i \cdot k)]$, and gauge invariance then forces $\sum_j q_j = \sum_i q_i$: **charge conservation**. The graviton version, one power of momentum higher, will force something much stronger — that all the couplings are *equal*. Keep the photon result in mind; the entire structure below is its copy with $\varepsilon_\mu \rightarrow \varepsilon_{\mu\nu}$ and $p^\mu \rightarrow p^\mu p^\nu$.

10 Emission from an initial-state leg [p. 23*****-25*****]

We build on our intuition given by the photon case.

We focus on the **emission factor** from an initial-state leg with momentum “ p_i ”



↪ The emitted graviton contributes with a polarization tensor $\varepsilon_{\mu\nu}^*(\vec{k})$.

↪ The vertex and the numerator of the propagator contribute with a rank-2 function of the four momenta involved: $\kappa_i^{\text{IN}} \Gamma^{\mu\nu}(p_i, k)$.

↪ Independently on the spin of the emitting particle, the denominator of the propagator gives the factor

$$\frac{1}{(p_i - k)^2 - m^2 + i\varepsilon} = \frac{1}{p_i^2 + k^2 - 2p_i \cdot k - m^2} = \frac{1}{-2p_i \cdot k} \quad \text{since } p_i^2 = m^2, k^2 = 0$$

↪ We indicate with κ_i^{IN} the coupling of the “i”-th particle in the initial state.

[p. 24*****]

In fact, in principle, each particle has its own coupling with the massless spin-2 emitted graviton. The emission factor has the form

$$\varepsilon_{\mu\nu}^*(\vec{k}) \frac{\kappa_i^{\text{IN}} \Gamma^{\mu\nu}(p_i, k)}{-2p_i \cdot k}$$

$\Gamma^{\mu\nu}$ is a polynomial in the four-momenta

In full generality, we write the generic structure

$$\begin{aligned} \Gamma^{\mu\nu}(p_i, k) = & g^{\mu\nu} A(p_i^2, p_i \cdot k) + 2p_i^\mu p_i^\nu B(p_i^2, p_i \cdot k) \\ & + (k^\mu p_i^\nu + k^\nu p_i^\mu) C(p_i^2, p_i \cdot k) + k^\mu k^\nu D(p_i^2, p_i \cdot k) \end{aligned}$$

↪ A, B, C, D are generic scalar functions

However, transversality $k_\mu \varepsilon^{\mu\nu}(\vec{k})^* = 0$ and the traceless condition $g_{\mu\nu} \varepsilon^{\mu\nu}(\vec{k})^* = 0$ imply that A, C and D do not contribute to the emission factor.

We thus have

$$\varepsilon_{\mu\nu}^*(\vec{k}) \kappa_i^{\text{IN}} \frac{2p_i^\mu p_i^\nu B(p_i^2, p_i \cdot k)}{-2p_i \cdot k} = -\kappa_i^{\text{IN}} \varepsilon_{\mu\nu}^*(\vec{k}) \frac{p_i^\mu p_i^\nu B(p_i^2, p_i \cdot k)}{p_i \cdot k}$$

The function $\Gamma^{\mu\nu}(p_i, k)$ has mass-dimension 2 (why?) and, therefore, the function B must be **dimensionless** [p. 25*****]

↪ Consequently, since $B(p_i^2 = m^2, p_i \cdot k)$ must be dimensionless, it can only depend on dimensionful quantities through dimensionless ratios, $B = B(p_i \cdot k/m^2)$

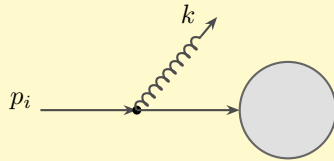
We thus find

$$-\kappa_i^{\text{IN}} \varepsilon_{\mu\nu}^*(\vec{k}) \frac{p_i^\mu p_i^\nu B(p_i \cdot k/m^2)}{p_i \cdot k} \approx -\kappa_i^{\text{IN}} \varepsilon_{\mu\nu}^*(\vec{k}) \frac{p_i^\mu p_i^\nu B(0)}{p_i \cdot k}$$

↪ In the soft approximation, we expand the function $B(p_i \cdot k/m^2)$ and the leading terms come from $B(0)$.

↪ Since $B(0)$ is a number, we can reabsorb it in a redefinition of κ_i^{IN} which is, at this stage, a free parameter.

We thus arrive at



$$\approx i\mathcal{M}_0(p_1, \dots, p_n; k_1, \dots, k_m) \left[-\kappa_i^{\text{IN}} \frac{\varepsilon_{\mu\nu}^*(\vec{k}) p_i^\mu p_i^\nu}{p_i \cdot k} \right]$$

Added explanation (not in the original notes)

Answering the “(why?)”. In four dimensions an amplitude with N external legs has mass dimension $4 - N$, so adding one external graviton lowers the dimension by one: the whole emission factor must carry dimension -1 . It equals $\kappa \Gamma^{\mu\nu} / (p_i \cdot k)$ with $[\kappa] = -1$ (comment 3) and $[\varepsilon] = 0$, so $[\Gamma] - [p_i \cdot k] = 0 \Rightarrow [\Gamma] = [M]^2$. Since $\Gamma^{\mu\nu}$ multiplies $2p_i^\mu p_i^\nu$ — already dimension 2 — the function B must be dimensionless. Concretely: $\Gamma^{\mu\nu}$ comes from the vertex $\kappa h_{\mu\nu} T^{\mu\nu}$ with $T^{\mu\nu} \sim \partial\Phi\partial\Phi$, i.e. two powers of momentum. Everything is consistent.

Why only B survives. A physical graviton polarization satisfies $k_\mu \varepsilon^{\mu\nu} = 0$ (transverse) and $g_{\mu\nu} \varepsilon^{\mu\nu} = 0$ (traceless) — precisely the TT-gauge conditions of p. 13^{*****}, now in momentum space. Contracting the four structures with $\varepsilon_{\mu\nu}^*$: the A term dies by tracelessness, the C and D terms die by transversality, and only $2p_i^\mu p_i^\nu B$ is left. *The two physical helicities of the graviton do all the filtering.*

Why the soft limit kills everything but $B(0)$. As $k \rightarrow 0$, $p_i \cdot k \rightarrow 0$, so $B(p_i \cdot k/m^2) \rightarrow B(0)$; the subleading terms of the Taylor expansion come with extra powers of k that cancel the pole and are therefore non-singular — *subleading* in the soft expansion. Only the pole term is universal, and its residue is $B(0)$, absorbed into κ_i^{IN} .

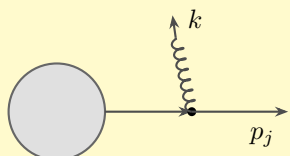
The structure to remember. Compare with the photon:

$$\text{photon: } -q_i \frac{\varepsilon_{\mu}^* p_i^\mu}{p_i \cdot k}, \quad \text{graviton: } -\kappa_i \frac{\varepsilon_{\mu\nu}^* p_i^\mu p_i^\nu}{p_i \cdot k}.$$

One power of p_i per index of the polarization. That extra power is what will turn charge conservation into the equivalence principle.

11 Emission from a final-state leg, and internal lines [p. 26^{*****}]

We have the same result, with a change of sign difference



$$\approx i\mathcal{M}_0(p_1, \dots, p_n; k_1, \dots, k_m) \left[+\kappa_j^{\text{FIN}} \frac{\varepsilon_{\mu\nu}^*(\vec{k}) p_j^\mu p_j^\nu}{p_j \cdot k} \right]$$

↪ The sign difference comes from the propagator denominator, since now

$$\frac{1}{(p_j + k)^2 - m^2 + i\varepsilon} = \frac{1}{2p_j \cdot k} \quad \text{compared to} \quad \frac{1}{-2p_i \cdot k}$$

of the previous computation.

■ Emissions from internal lines

All sub-leading, as in the photon case.

Added explanation (not in the original notes)

Where the sign comes from, in one line. For an *incoming* leg the internal propagator carries $p_i - k$; for an *outgoing* leg it carries $p_j + k$. Expanding, $(p_j + k)^2 - m^2 = +2p_j \cdot k$ versus $(p_i - k)^2 - m^2 = -2p_i \cdot k$. Hence the relative minus sign between the two sums in the theorem.

Why internal emissions are subleading. A graviton attached to an internal line does not produce a propagator that goes on-shell as $k \rightarrow 0$: the internal momentum is already off-shell by a finite amount, so the extra propagator stays finite and the contribution has *no* $1/(p \cdot k)$ pole. Only external legs can go nearly on-shell, and only they produce the soft divergence.

A notation slip in the original. On this sheet the final-state momenta are written p_j , but everywhere else — including the theorem on the next page — they are called $k_1, \dots, k_m$. Read $p_j \equiv k_j$ here; nothing else changes.

12 The soft theorem and gauge invariance [p. 27*****]

We now discuss the theorem.

In the soft limit, the amplitude for the emission of a soft graviton takes the form:

$$i\mathcal{M}_1(p_1, \dots, p_n; k_1, \dots, k_m; k) \approx i\mathcal{M}_0(p_1, \dots, p_n; k_1, \dots, k_m) \times \\ \varepsilon_{\mu\nu}^*(\vec{k}) \left[\sum_{j=1}^m \kappa_j^{\text{FIN}} \frac{k_j^\mu k_j^\nu}{k_j \cdot k} - \sum_{i=1}^n \kappa_i^{\text{IN}} \frac{p_i^\mu p_i^\nu}{p_i \cdot k} \right]$$

$\hookrightarrow$ The physics must remain invariant if we consider instead of $\varepsilon_{\mu\nu}(\vec{k})$ the polarization tensor

$$\varepsilon_{\mu\nu}(\vec{k}) + \xi_\mu k_\nu + \xi_\nu k_\mu$$

Consequently, if we substitute $\varepsilon_{\mu\nu}^*(\vec{k})$ with either k_μ or k_ν , we must get a vanishing amplitude.

We thus write

$$k_\mu \left[\sum_{j=1}^m \kappa_j^{\text{FIN}} \frac{k_j^\mu k_j^\nu}{k_j \cdot k} - \sum_{i=1}^n \kappa_i^{\text{IN}} \frac{p_i^\mu p_i^\nu}{p_i \cdot k} \right] = \\ = \sum_{j=1}^m \kappa_j^{\text{FIN}} \frac{\cancel{(k \cdot k_j)} k_j^\nu}{\cancel{k_j \cdot k}} - \sum_{i=1}^n \kappa_i^{\text{IN}} \frac{\cancel{(k \cdot p_i)} p_i^\nu}{\cancel{p_i \cdot k}} \stackrel{!}{=} 0$$

We thus find

$$\sum_{j=1}^m \kappa_j^{\text{FIN}} k_j^\mu = \sum_{i=1}^n \kappa_i^{\text{IN}} p_i^\mu$$



Added explanation (not in the original notes)

Why $\varepsilon_{\mu\nu} \rightarrow \varepsilon_{\mu\nu} + \xi_\mu k_\nu + \xi_\nu k_\mu$. This is the gauge transformation $h^{\mu\nu} \rightarrow h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$ written in momentum space: for a plane wave $h^{\mu\nu} \sim \varepsilon^{\mu\nu} e^{-ikx}$, the derivative becomes $-ik$. Since the shift must not change any physical amplitude, replacing $\varepsilon_{\mu\nu}^*$ by $k_\mu \xi_\nu + k_\nu \xi_\mu$ must annihilate the amplitude for arbitrary ξ — which is the condition used above. This is the

graviton **Ward identity**, the exact analogue of $k_\mu \mathcal{M}^\mu = 0$ for photons (Lecture 11).

The cancellation. Contracting k_μ with $k_j^\mu k_j^\nu$ gives $(k \cdot k_j) k_j^\nu$, and the propagator denominator is exactly $k_j \cdot k$ — so the pole cancels and each term collapses to $\kappa_j k_j^\nu$. The dangerous soft singularity disappears, leaving a purely kinematic condition on the couplings.

13 Gravity is universal: the equivalence principle [p. 28*****]

The point is that the four momenta already satisfy the condition

$$p_1^\mu + \cdots + p_n^\mu = k_1^\mu + \cdots + k_m^\mu$$

★★

The only way to make ★ and ★★ compatible is to take all couplings equal

$$\kappa_i^{\text{IN}} = \kappa_j^{\text{FIN}} \equiv \kappa, \quad \forall i, j$$

Gravity is universal (the equivalence principle)

Added explanation (not in the original notes)

Why “the only way”. Condition ★ says $\sum_j \kappa_j^{\text{FIN}} k_j^\mu = \sum_i \kappa_i^{\text{IN}} p_i^\mu$ for all kinematic configurations, while momentum conservation ★★ says $\sum_j k_j^\mu = \sum_i p_i^\mu$. The momenta are otherwise independent (one can scatter any set of particles at any angles), so the only way a weighted sum can equal an unweighted one identically is for all the weights to coincide. Then ★ is nothing but $\kappa \times \text{★★}$ and is automatically satisfied.

This is a derivation of the equivalence principle. We did not *assume* that all bodies fall alike; we assumed only Lorentz invariance, gauge invariance of a massless spin-2 field, and momentum conservation. Universality of the gravitational coupling came out. Compare with the photon, where the same argument gives $\sum_j q_j = \sum_i q_i$ — charge conservation, but individual charges may still differ, which is why the electron and the proton can have opposite charges while falling identically in a gravitational field.

And one step further (Weinberg’s punchline, not in the notes). Repeating the argument for a massless particle of spin $s \geq 3$ would require $\sum_j \kappa_j k_j^{\mu_1} \cdots k_j^{\mu_{s-1}} = \sum_i \kappa_i p_i^{\mu_1} \cdots p_i^{\mu_{s-1}}$, a condition with no non-trivial solution. *Massless higher-spin particles cannot have long-range interactions.* Spin 1 gives charge conservation, spin 2 gives the equivalence principle, and there the list stops.

14 The graviton propagator [p. 28*****]

Consider the Lagrangian

$$\begin{aligned} \mathcal{L}_0 = & \frac{1}{2} (\partial_\mu h_{\rho\sigma}) (\partial^\mu h^{\rho\sigma}) - (\partial_\mu h_{\nu\rho}) (\partial^\nu h^{\mu\rho}) + (\partial_\mu h^{\mu\nu}) (\partial_\nu h) \\ & - \frac{1}{2} (\partial_\mu h) (\partial^\mu h) \end{aligned}$$

↔ This Lagrangian has a gauge invariance $h^{\mu\nu} \rightarrow h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$ and we cannot derive the propagator without fixing the gauge.

The situation is conceptually analogue to that of QED. The Lagrangian $-\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$ is gauge invariant and, to extract the propagator, we need...

[The sheet, and with it the scan, breaks off here in mid-sentence.]

Added explanation (not in the original notes)

Completing the broken sentence. “...we need to fix the gauge, e.g. by adding a gauge-fixing term $-\frac{1}{2\zeta}(\partial_\mu A^\mu)^2$ to the Lagrangian”. The reason is purely technical but unavoidable: a gauge-invariant quadratic form is *degenerate*. Its kinetic operator annihilates every pure-gauge configuration $\partial^\mu \alpha$ (respectively $\partial^\mu \xi^\nu + \partial^\nu \xi^\mu$), so it has zero modes, so it cannot be inverted — and the propagator is the inverse of the kinetic operator. Adding a gauge-fixing term removes the zero modes and makes inversion possible; physical results do not depend on the gauge parameter.

Where this is going. In QED, gauge fixing with parameter ζ gives the photon propagator $\frac{-i}{k^2 + i\epsilon} [g^{\mu\nu} - (1 - \zeta) \frac{k^\mu k^\nu}{k^2}]$, and $\zeta = 1$ (Feynman gauge) makes it simply $-ig^{\mu\nu}/k^2$. The analogous procedure for $\mathcal{L}_0$ — the harmonic/de Donder gauge $\partial_\mu h^{\mu\nu} = \frac{1}{2} \partial^\nu h$ — gives the graviton propagator

$$\frac{i}{2} \frac{g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho} - g^{\mu\nu} g^{\rho\sigma}}{k^2 + i\epsilon},$$

whose numerator structure (in particular the $-g^{\mu\nu} g^{\rho\sigma}$ piece) is what makes gravity *attractive* between like sources, in contrast to electromagnetism. Presumably the next lecture does exactly this.

Summary of Lecture 17 — the logic in eight lines

1. In the massless limit the Fierz–Pauli mass term dies (it is not gauge invariant) but the kinetic part survives, and the field acquires the gauge invariance $h^{\mu\nu} \rightarrow h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$.
2. The free E.o.M. is $\mathcal{G}^{\rho\sigma}(h) = 0$, where $\mathcal{G}$ is twice the linearized Einstein tensor and satisfies the off-shell identity $\partial_\rho \mathcal{G}^{\rho\sigma} \equiv 0$ (linearized Bianchi).
3. A gauge choice reduces $h^{\mu\nu}$ to the TT gauge $h^{\mu 0} = \partial_i h^{ij} = h = 0$ — two physical DOF, exactly as Coulomb gauge does for the photon.
4. Interactions: add $-\kappa h_{\mu\nu} T^{\mu\nu}$. Gauge invariance forces $T^{\mu\nu}$ to be *symmetric and conserved*; the E.o.M. $\mathcal{G}^{\rho\sigma} = -\kappa T^{\rho\sigma}$ is the linearized Einstein equation, and it re-derives $\partial_\mu T^{\mu\nu} = 0$ by itself.
5. The *canonical* $T^{\mu\nu}$ is generally not symmetric when the fields carry spin: $\partial_\mu S^{\mu\rho\sigma} = T^{\sigma\rho} - T^{\rho\sigma}$.
6. The Noether ambiguity $\Theta^{\mu\nu} = T^{\mu\nu} + \partial_\lambda K^{\lambda\mu\nu}$ with K antisymmetric in its first two indices preserves conservation and the charges, and can be used to symmetrize: $K^{\lambda\mu\nu} = \frac{1}{2}(S^{\lambda\mu\nu} + S^{\mu\nu\lambda} - S^{\nu\lambda\mu})$ — the **Belinfante–Rosenfeld** tensor.
7. $[\kappa] = [M]^{-1}$: linearized gravity is non-renormalizable, an effective theory valid below M_{Pl} .
8. Soft-graviton emission is universal, $\propto \varepsilon_{\mu\nu}^* p^\mu p^\nu / (p \cdot k)$; gauge invariance plus momentum conservation then force *all* couplings to be equal — **gravity is universal**, which is the equivalence principle.